# PM: Doped Inert Single Crystals for Probing Beyond the Standard Model

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One of the outstanding puzzles in fundamental physics has to do with the matter/antimatter asymmetry in the universe. An ingredient needed to explain this asymmetry is CP (Charge-Parity) violation; however the amount of CP violation currently accounted for in the standard model of particle physics is insufficient. The search for electric dipole moments (EDMs) of fundamental particles is a probe for new sources of CP violation needed to allow for our existence.

Because of their large internal electric fields, molecules probed by precision spectroscopy offer a promising path for discovering EDMs not predicted by the standard model. The most precise searches for the electron EDM have recently been performed on molecules held in an ion trap [1] or in a molecular beam [2]. Next-generation EDM experiments seek to gain better statistical sensitivity, both for the intrinsic value of reducing statistical uncertainty and because more powerful statistics are crucial for understanding and reducing sources of systematic uncertainty.

Atoms and molecules gently encased in the 'artificial vacuum' of an inert cryogenic crystal matrix [3, 4, 5, 6, 7, 8, 9, 10, 11] have been proposed as promising candidates for greatly improved statistical sensitivity over what can be achieved in traps and beams. Potential applications include search for static EDMs discussed above, as well as time-varying EDMs (e.g. due to dark matter), other ultralight dark matter signatures, nanoscale NMR, and single-molecule NMR. The EDM$^3$ collaboration is currently developing an electron EDM experiment using BaF molecules frozen into a polycrystalline rare gas matrix [10]. The advantage of this approach over previous solid-state atomic EDM experiments is three-fold: (1) the well-known molecular enhancement factors as compared with most atoms, (2) the potential to use optical techniques for state preparation and readout, and (3) the achievable purity of cryogenic rare gas host crystals as compared with most room temperature host matrices.

These efforts are in the early stages of development, but they show great potential. Dopants could be loaded into a cryocrystal with low enough density (spacing $\sim 30$ nm) for inter-dopant magnetic interactions to still allow a 1 ms electron spin coherence time, similar to the transit time in the ACME-II [2] experiment which produced the leading electron EDM value. That dopant density would provide a huge statistical sensitivity, such that a doped cryocrystal could in principle provide in seconds the same uncertainty obtained by ACME in their 350-hour data set. Said a different way, the doped crystal would have over $10^{13}$ molecules in a 1 mm$^3$ volume, which can be compared with the few hundred thousand molecules in the ACME-II beam at any given time.

But the potential advantages offered by doped cryocrystals can only be realized if the dopants retain their crucial properties: the ability to efficiently prepare and measure the quantum spin state, the ability to perform coherent operations on those states, and long coherence times. Using the same optical techniques employed in gas-phase experiments, these properties have been demonstrated for atoms trapped in solid helium [12, 13, 14, 15], in parahydrogen [4, 16, 17, 18], and in neon [19]. Key properties have been demonstrated, but with performance limited by the polycrystalline structure of the grown cryocrystals [16, 17].

The many possible local configurations (so-called 'trapping sites') of dopants within a polycrystalline host matrix causes the ensemble spin dephasing time to be orders of magnitude shorter than the spin echo coherence time [17, 18], demonstrating that inhomogeneous broadening could limit many applications of

| System | Milestones (Year 1-3) | 1 | 2 | 3 |
|---|---|---|---|---|
| Ar crystals grown on semi-crystalline substrate | X-ray characterization of undoped and doped crystals | ■ | | |
| | Optical analysis of doped Cs, including $T_1$, $T_2^*$ | ■ | | |
| Ar crystals grown on single crystal Ar substrate | X-ray characterization of undoped and doped crystals | | ■ | |
| | Optical analysis of doped Cs, including $T_1$, $T_2^*$ | | ■ | |
| | Explore growth parameters to obtain good $T_1$, $T_2^*$ up to mm scale | | ■ | |
| | Achieve density-$T_2^*$ product of 100x previous record | | ■ | ■ |
| Doped Ne crystals | Similar x-ray and optical analysis | | | ■ |

Table 1: Summary timeline of selected milestones.

doped cryocrystals. The creation of doped single crystals will be needed to unlock the full potential. To date, the vapor-deposition techniques used to grow matrices with high densities of dopants have resulted in polycrystalline samples [20, 21, 22]. However, it was demonstrated decades ago that obtaining single rare-gas crystals was possible if they were grown at the liquid-solid interface [23, 24, 25, 26, 27, 28]. This process is analogous to how large semiconductor crystals are 'pulled' from the liquid. However, nobody has found a way to implant high densities of dopants with the latter technique.

**In this work, we will develop a new technique to grow single-crystal hosts with high densities of dopants**. We will grow an undoped single crystal from the melt, as has previously been demonstrated. We will use this single crystal as a seed for subsequent vapor deposition of a doped crystal. Vapor deposition is routinely used to grow 2D single crystals, so this technique appears promising.

In recent years, most experimenters working with doped cryocrystals have relied on FTIR and optical techniques to infer properties of the crystal structure and the nature of dopant trapping sites. This proposal, however, puts a premium on obtaining a single-crystal host. Therefore we will rely heavily on more direct x-ray analysis to study the properties of the host crystal. We will also use x-ray techniques to infer properties of the dopant trapping sites. Finally, once we are satisfied with what the x-ray analysis says about our doped crystals, we will use optical techniques to determine whether the properties of the dopants are living up to their promise for precision measurements.

# 1 Properties of Dopants Trapped in Inert Matrices

Rare gas crystals, formed by weak van der Waals interactions, can be expected to represent the most non-perturbative hosts for dopants. Even when polycrystalline, when these matrices are of reasonable quality, they are plenty transparent for optical manipulation and probing of alkali atoms. However, with inert crystal lattice spacings of a few to several angstroms, some properties of dopant atoms and molecules are quite different from those in vacuum. Alkali atom excited state lifetimes in inert matrices have been measured to be comparable with their gas-phase values, indicating that radiative decay is still the dominant pathway [29, 30]. Absorption and fluorescence have often been used to read out the spin state of alkali atoms, but the emitted spectrum is broadened some tens of nanometers, and red-shifted tens to hundreds of nanometers [31, 29, 30]. (We should note that an M1 transition of thulium atoms trapped in rare gas matrices have been measured to have sub-nm linewidth, suggesting interesting uses for these types of transitions [11].) Optical pumping of the electron spin state has been demonstrated with varying degrees of efficiency [12, 14, 32]. Alkali atoms in inert matrices typically 'bleach', i.e. stop scattering photons after some large number of cycles, but this is sometimes reversible by application of different wavelength light [30]. Bleaching is potentially a concern for precision measurements, and it will be discussed in Sec. 5.

For an EDM measurement, we are ultimately concerned with the *ensemble* transverse spin relaxation

time $T_2^*$ achievable for samples with a high density of dopants. The longitudinal spin relaxation time $T_1$ and the transverse spin relaxation time $T_2$ (measured using some spin echo technique to mitigate the effects of inhomogeneous broadening) are relevant because they set upper limits on $T_2^*$. Early work on Cs in helium matrices obtained $T_2^* = 0.1$ s, which demonstrated the great promise of doped cryocrystals for precision measurements. But these results were limited to low dopant densities associated with challenges of injecting dopants into solid helium [13, 14, 15]. These issues will be discussed in Sec. 2.

When considering dopant densities sufficiently high to be of interest for EDM experiments, the results of the Weinstein group on Rb atoms in polycrystalline parahydrogen [17, 18] and neon [19] matrices serve as the most relevant benchmarks. For Rb in parahydrogen, they measured values up to $T_1 = 0.2$ s and $T_2 = 0.1$ s, but the best-case $T_2^*$ was $\sim 100$ $\mu$s. For Rb in neon, they obtained up to $T_1 = 0.6$ s and $T_2 = 0.1$ s, but the best-case $T_2^*$ was $\sim 10$ $\mu$s. The critical question is what type of inhomogeneities account for the orders of magnitude gap between $T_2$ and $T_2^*$. In both parahydrogen and neon (as well as for Cs in parahydrogen [17]) it was determined that, while magnetic-like effects limited $T_2$, electrostatic-like effects limited $T_2^*$. In other words, $T_2$ is limited by magnetic impurities, and $T_2^*$ is limited by inhomogeneous electrostatic interactions with the local crystal field–caused by Cs in a large number of different trapping sites.

What does all this mean for an EDM experiment in a cryogenic matrix? For magnetic sensing applications, spin-echo techniques can be employed, so $T_2$ is the relevant quantity. However, there is no known way to use spin echo techniques for an EDM measurement, so the ensemble coherence time $T_2^*$ sets the limit on experimental spin-precession times. In a polycrystalline environment, having a large degree of trapping site inhomogeneity would be expected, both because of the different crystal structures (mix of fcc and hcp) [20, 33, 34, 35] and orientations of the microcrystallites, and because of the often-observed tendency of dopants to accumulate at grain boundaries. For doped cryocrystal experiments to obtain their full imagined potential, a way must be found to dramatically reduce trapping site inhomogeneity. It is our conjecture, to be tested in this proposal, that achieving a single-crystal host with only a single type of dopant trapping site will enable high-dopant-density samples with $T_2^*$ long enough to be of great interest for EDM experiments.

## 2 Choices of Species

The goals of the project are

- Produce large undoped inert crystals in a manner consistent with introducing dopants
- Produce doped crystals exhibiting only a single type of dopant trapping site
- Make the first demonstration of long electron dephasing times for high densities of dopants

In this section we discuss choices for host and dopants for achieving these goals.

### a. Choice of Dopant

Ultimately, we imagine an EDM experiment using polar molecules trapped in a cryogenic matrix, various forms of which have been proposed by Kozlov and Derevianko [6, 7] and the Vutha, Horbatsch, and Hessels of the EDM$^3$ collaboration [9, 10]. However, this proposal is concerned with the foundations which will enable those experiments. As such, we will work with the obvious candidate for dopants: alkali atoms. On one hand, the open-shell electron with its $p$ excited state is expected to have potentially problematic interactions with the host crystal. But an alkali atom is still a good choice, because these problems would also be present for a molecular radical usable for an electron EDM experiment.

Our dopants will be Cs atoms. Cs in parahydrogen polycrystals has a demonstrated electron spin ensemble dephasing time $T_2^* \approx 1$ $\mu$s [17, 16]. A single-crystal with a well-defined local dopant configuration

should eliminate electrostatic inhomogeneous broadening, enabling a $T_2^*$ to approach the $T_2 > 1$ ms observed in spin-echo experiments in polycrystalline samples (at high atomic densities of $10^{16}$ cm$^{-3}$) [18]. For dopant molecules or non-$s$ ground state atoms, where the inhomogeneous broadening from electrostatic-like interactions with the host matrix would be expected to be greater, the gains over the current state-of-the-art could be even larger than the factor of $10^3$ expected for Cs in parahydrogen.

For the work of this proposal, there are a few reasons to choose Cs over the other alkalis.

- X-Ray analysis. Cs is the heaviest of the stable alkalis, so X-ray analysis of the local dopant configurations will yield more information than for the lighter alkalis.

- Optical pumping and readout. The ability to optically pump and read out the electron spin state has been found to generally favor atoms with a large fine structure splitting. This trend makes sense, since photon angular momentum couples to the orbital angular momentum of the electron, which is only coupled to the electron spin through spin-orbit coupling. Electrostatic coupling of the electron position to the host matrix results in a decoupling of its spin and orbital degree of freedom. The fine-structure splitting in the excited state of Cs is 554 cm$^{-1}$, compared with 237 cm$^{-1}$ for Rb, and 57 cm$^{-1}$ for K.

- Sensitivity to trapping site configuration. Atoms with larger hyperfine constants have generally exhibited smaller $T_2^*$ times. This is understood to be as a result of electrostatic interactions with the host crystal. Tensor Stark shifts of $s$-state alkali atoms arise at third-order perturbation theory, with the Stark Hamiltonian in second-order and the hyperfine interaction Hamiltonian in first-order and proportional to the hyperfine constant [36, 37]. So, the choice of Cs might not be best for obtaining record $T_2^*$ times. However, it gives us the largest diagnostic handle on the effects we are trying to eliminate for other species in future EDM experiments.

- Size. The ultimate EDM molecules will almost certainly involve a heavy atom and an unpaired electron, meaning that it will be big. The large size of Cs is well matched for that ultimate goal.

### b. Choice of Host Crystal

Solid helium appears at first to be a promising candidate for the host. Of the inert gases, helium is the least polarizable so will be the least interactive. It has a high-symmetry bcc phase in its ground state. And it is a quantum solid, meaning that the wavefunction spread is of order the lattice spacing, and thus the lattice is predicted to rearrange itself relatively freely about each dopant atom. Indeed, the best behavior of dopants has been obtained in solid helium. Weis and collaborators showed Cs dopants could exhibit $T_1$ times as long as 1 s [12] and $T_2^* = 0.1$ $s$ [13]. Unfortunately, solidifying helium requires working at high pressures, and this has caused a problem for doping. The Weis group used laser ablation to implant Cs clusters into the solid, some of which were broken up into atoms with a subsequent ablation pulse. They obtained cluster densities of $\sim 10^{10}$/cm$^3$ and single-atom densities of $\sim 10^9$/cm$^3$. Looking forward to future EDM experiments with polar molecules, that dopant density is not so different from what can be obtained in molecular beams, so this technique would not appear to be competitive with other EDM experiments. Also, the high cluster density left over by this technique could be expected to cause a problem for EDM experiments using molecular radicals.

All the noble gas crystals have comparable nearest-neighbor spacings of $3.7 - 4.4$ Å. Helium has the lowest polarizability, $\alpha = 0.21$ Å$^3$. Neon has the next lowest polarizability, $\alpha = 0.40$ Å$^3$, argon has $\alpha = 1.63$ Å$^3$, and xenon has the highest, $\alpha = 4.02$ Å$^3$ [15]. In this proposal, we will work with first argon and then neon host crystals. For initial work, we desire a heavy-nucleus host to give larger x-ray analysis signals. Xenon would be best in this regard, but it's $\sim 50\%$ abundance of $I \neq 0$ isotopes means that Cs $T_2^*$

times would be limited to $\sim$ 10 $\mu$s. Similarly, naturally occurring krypton has a problematic abundance of an $I = 9/2$ isotope. Isotopically enriched gases could be used, but instead we will work with argon, which has only $I = 0$ naturally occurring isotopes, for our heaviest host crystal. Neon is also worth exploring because its low polarizability might be advantageous for obtaining the best $T_2^*$ values. However, the 0.3% abundance of an $I = 3/2$ isotope would limit the Cs $T_2^*$ to $\sim$ 1 ms, so we might need to work with isotopically enriched samples after demonstrating initial good results. Also, neon's light mass will require work at Northwestern's dedicated DND-CAT facility at Argonne National Lab, rather than our on-campus x-ray facility.

Early work observed the absorption spectra of Cs trapped in neon and argon matrices [38, 39]. This project would appear to represent the first study of optical pumping and spin relaxation processes for these dopant/host combination.

## 3 Growth Techniques

Nobody to date has obtained high dopant densities in a high-quality inert gas single crystal host. The vapor-deposition techniques used to grow matrices with high dopant densities have resulted in polycrystalline samples [20, 21, 22]. However, growing from the liquid phase ('from the melt', in a similar manner as very large semiconductor crystals are grown) has yielded centimeter-sized single crystals of inert gase atoms [27, 23, 24, 40, 24, 25, 26].

The work of Hänsch and Weis using laser ablation to inject Cs atoms into solid helium has already been discussed, along with its limitations. Yabuzaki and collaborators implanted Rb, Cs, Ba, and Ca atoms into liquid helium through 'laser sputtering' of solid metal inside the liquid helium [41]. With this laser beam alone no atoms were seen, but a secondary pulsed beam in the liquid dissociated clusters and atomic LIF was observed. They estimate an atomic density of $10^8$ to $10^{10}$ per cc steady-state. Furukawa et. al. [42] dissociated the clusters in the liquid with a femtosecond pulsed laser. One might be able to use such techniques to implant dopants in the liquid, and then grow a high-quality doped crystal from the melt. However, the tendency of dopants to find one another and cluster would be a concern, both because of depletion of the desired dopant densities and because of contaminating effects of the clusters.

Growing doped single crystals by vapor deposition seems a promising approach. Nobody has succeeded at this yet, but neither has anyone tried the techniques we propose here. Single 2D crystals up to the hundred-micron scale can be grown by vapor deposition. This demonstrates that atoms tend to find the lowest-energy configurations as they deposit. In these experiments, the limitations on grain size arise from contaminating impurities and from multiple non-commensurate seeding sites. It turns out that lattice matching between the substrate and the deposited layer is not important, because the 2D crystal forms with relative lattice vector at an angle which minimizes the energy. However, having a substrate which is not quite but nearly atomically flat is extremely important [43].

When growing from the melt, single crystals of inert gas atoms have been grown with apparently no concern whatsoever for the substrate. This can be achieved because atoms in the liquid have high enough mobility to heal any defects at the surface, made significantly easier for the inert gases because they only interact weakly with the surface through van der Waals forces. Our key idea is to use a large-area undoped single crystal, grown initially from the melt, *as a substrate* for subsequent growth by vapor deposition of the doped crystal. In this technique, surface flatness of the undoped seed crystal will be a non-issue. This technique does require more complicated cryogenic engineering as compared with only doing vapor deposition, but that is a surmountable challenge. During the preparation of this proposal, we learned from Eric Hessels that the EDM$^3$ collaboration had arrived at a similar idea (before we did) and are planning to pursue it. However, they have no current plans to test the idea first with atoms as dopants, nor so they plan to perform x-ray crystallography.

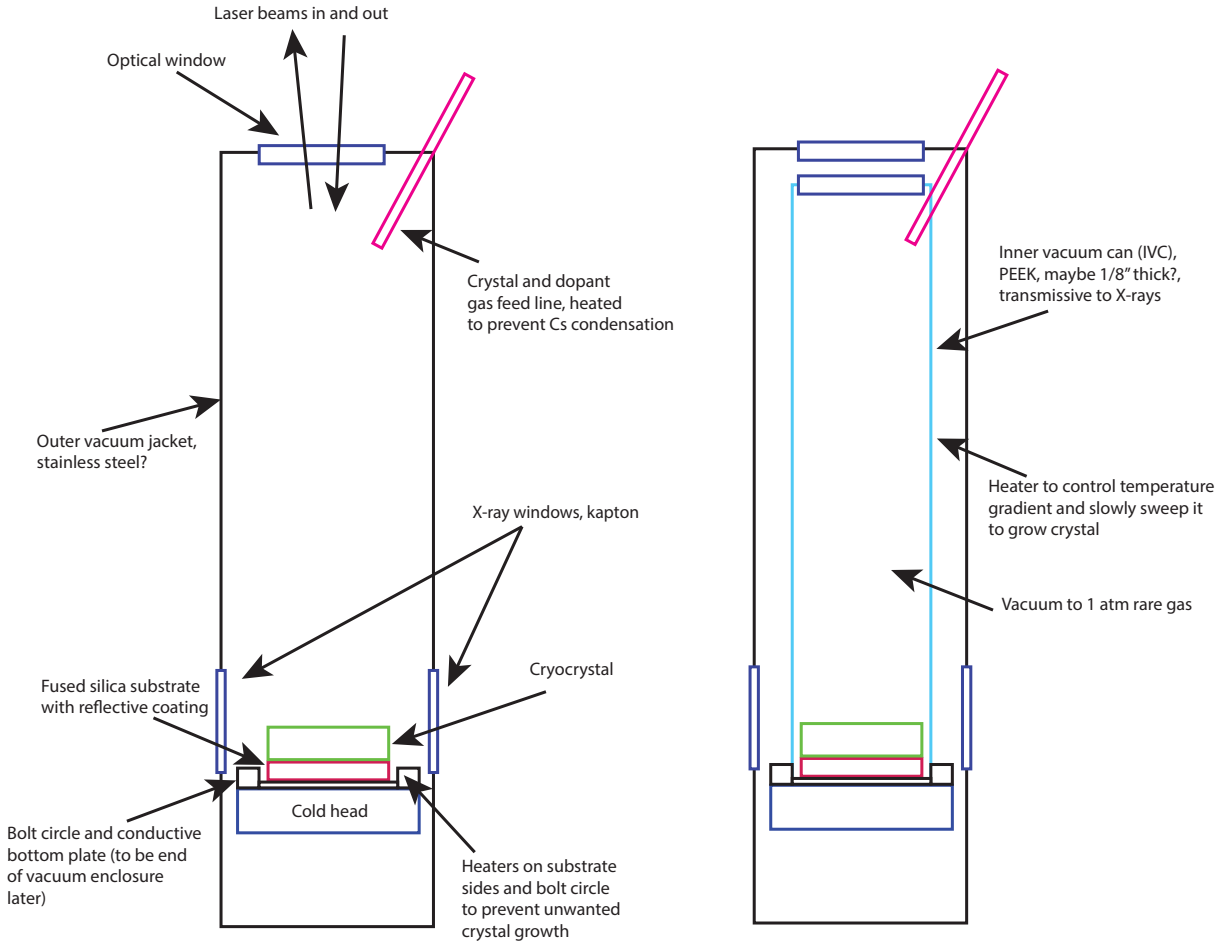



Figure 1: Cryostat assembly in Phase I and Phase II. Left: Phase I, for an easy start, but relatively small crystal domains. Right: Phase II, for obtaining mm- or cm-scale single crystals by sweeping liquid-solid interface and possibly annealing. X-ray analysis will be used in both phases, without and with dopants. Optical analysis will be used in both phases when dopants are present.

### a. Getting Started with an Easy Substrate

The group of our collaborator Michael Bedzyk can grow 2D mica and silver crystals on various substrates including commercially available single-crystal quartz and fused silica. These 2D crystals will form a high-quality reflective surface for our optical analysis of dopants. The grain sizes are typically 5 nm. These relatively small grain sizes will certainly lead to polycrystalline samples when growing doped crystals directly onto them by vapor deposition. However, vapor deposition of doped crystals directly onto these surfaces can be done in a less complicated apparatus, and will serve as a testbed for developing our technical skills. If it becomes of interest, Jiwoong Park of the University of Chicago Department of Chemistry has offered to provide us with *atomically flat* reflective 2D crystals with grain sizes of $\sim 100\ \mu$m.

Fig. 1 shows the two phases of the apparatus. The first is designed to limit complexity while we are working with polycrystalline doped samples grown directly reflective substrates by vapor deposition. Vapor deposition can be done in this apparatus by operating at pressures below the triple point, for which gas deposits directly on the cold substrate without going through a liquid phase. Operation at low pressures in this apparatus is critical to prevent too much of a heat load on the cryostat. Once we have the various techniques under control, we will upgrade the apparatus to the double-enclosure version, which will allow creation of a single-undoped-crystal seed for subsequent doped crystal growth.

### b. Growing Onto a Single-Crystal Substrate

The double-enclosure apparatus in Fig. 1 will allow the relatively high pressures required to operate above the triple point. Thereby, we will gain the ability to work with a liquid phase and grow single undoped crystals from the melt, for use as seeds of the doped crystals later grown by vapor deposition. This inner chamber is equipped with various heaters and thermometers which will be necessary for the process.

We will first fill the cell with the noble gas, and then lower the temperature of the cold head to condense liquid on the substrate at the bottom of the cell. Once the desired amount of liquid is accumulated, maybe $1-10$ mm above the substrate, we will slowly lower the temperature of the cold head further so that a solid begins to form on the substrate. (This temperature sweep is the equivalent of process by which large-volume single crystals of semiconductor are 'pulled' from the melt.) The apparatus is equipped with various heaters designed to ensure that crystallization is only occurring on the substrate. This is an important point, because if the solid filled up the entire width of the inner chamber, there would be unwanted strains on the crystal structure as the temperature and pressure is changed.

Once the undoped seed crystal is formed, the pressure is slowly lowered. The pressure will always remain high enough so as to not have significant sublimation from the crystal. (Depending on the temperature, this could be accomplished at a pressure either above or below the triple point. It will probably be desirable to operate below the triple point pressure because of the subsequent vapor deposition phase.) But the pressure will be low enough so that the other parts of the cell, which have are being heated to elevated temperatures, boil off their liquid.

After the liquid has boiled off, we will be left with only gas and an undoped single crystal. At this point, more of the same noble gas will be let in from the top of the chamber, this time entrained with Cs atoms carried along by hydrodynamic transport. The inlet nozzle will be heated to prevent condensation of either noble gas or Cs.

This protocol requires a large temperature gradient along the height of the inner vessel, with temperatures as low as 4 K at the cold head and slightly above room temperature at the nozzle. For this reason, the inner vessel will be constructed from a thermal insulator such as PEEK, to avoid too much of a thermal load on the cryostat. A PEEK vessel with 1/8" wall thickness has a low enough thermal conductivity to meet these demands.

### c. Annealing

The double enclosure setup will also allow us to experiment with a wide range of annealing parameters, again without putting too much of a load on the cryostat. Annealing of undoped cryocrystals has been shown to have some success [44, 22, 45]. Work with doped crystals has not been done in an environment like we propose here, so the range of annealing temperatures and pressures was less than desired. It is an open question whether some range of annealing parameters can be found which allow the matrix to relax into a more perfect crystal, without providing the dopants with enough mobility to cluster or migrate to the edges of the host.

With the right combination of gas pressure and heater settings, we could obtain a scenario where the growth-surface of the crystal is elevated in temperature compared with the bulk, which is closer to the cold head. This arrangement could enhance mobility to the dopants, some level of which is desirable for arriving at the single lowest-energy local configurations. Too much mobility is undesirable because it would be expected to lead to dopant clustering. Laser excitation of alkali dopants can lead to undesirable bleaching (discussed in Sec. 5), by providing the atoms enough energy to reconfigure their local environment. However, during crystal growth it is possible that this would be beneficial, and could achieve a sort of continuous annealing as the crystal is grown. We will investigate these and other possibilities.

## 4 X-ray Analysis

In the 1960s, X-ray diffraction (XRD) was used to measure the lattice constant, thermal expansion, and isothermal compressibility of argon and neon single crystals ($\sim$3 mm diameter $\times$ $\sim$25 mm long) [27, 28]. The XRD technique used in those experiments employed a film technique with an oscillating back-reflection x-ray camera.

Relative to those early single-crystal XRD studies, we will use upgraded techniques. Each of the two facilities we will use is equipped with a heavy-duty high-performance Huber 4- or 6-circle diffractometer with a point detector or a modern area detector. Our cryostat and vacuum enclosure are designed to be compatible with these machines. Referring to Fig. 3, the mode of operation is to have a fixed incident beam direction and rotate the cryostat by orthogonal rotational axes $\phi$, $\chi$, and $\omega$, allowing for arbitrary sample orientation relative to the beam. The detector rotates in 2 (co-axial with $\omega$) to vary the scattering vector. Advantages of this setup over that used in early work are a wider angular range to align a larger selection of *hkl* Bragg reflections, (where *hkl* refer to the indices of the lattice planes), higher angular precision for lattice parameter determination, single photon counting for low background, and computer controlled stepping motors and detector counters for automation of data collection. Also we will be able to use x-rays to study the dopant configurations and unwanted impurities.

### a. X-Ray Expertise at Northwestern

This proposal has the strong support of two (unfunded) Northwestern University collaborators. We will rely on their expertise for the x-ray analysis, so it is important to give some of their credentials here.

- **Michael Bedzyk** is an NU Professor of Materials Science and Engineering and has a courtesy appointment in the Department of Physics and Astronomy. He is a physicist by training and a Fellow of the American Physical Society. His research program involves X-ray characterization of materials down to atomic resolution. He is the NU's Faculty Director of the X-Ray Diffraction Facility and co-director of the Synchrotron Research Center. He has a longstanding (up to this point non-professional) interest in fundamental symmetries, and is thus motivated by the goals of this proposal.

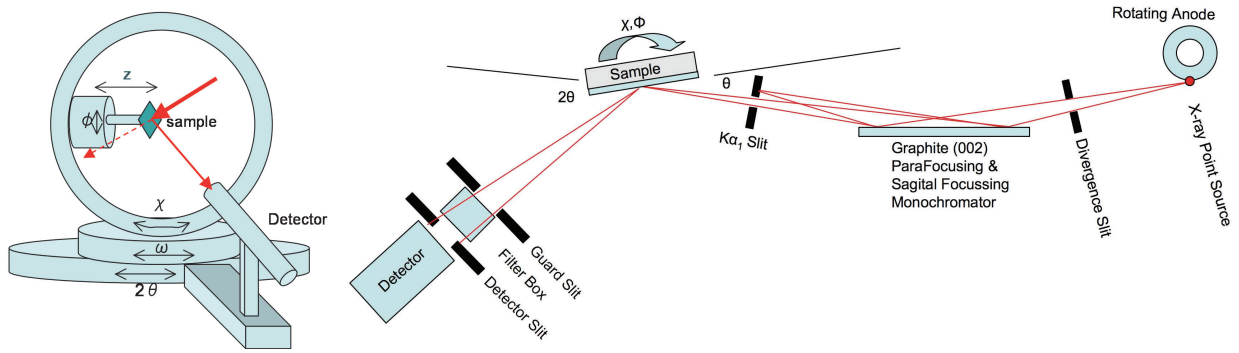



Figure 2: Schematics for NU XRD Facility Huber 4-circle diffractometer showing Cu rotating anode X-ray source followed by graphite doubly focusing monochromator and an incident beam slit that only passes 8.041 keV Cu K$\alpha_1$ X-rays to the sample with. wavelength $\lambda = 1.541\mathring{A}$. The incident and diffracted beams lie in the horizontal plane for this setup. The $\omega$, $\chi$, and $\phi$ axes can rotate the selected $G_{hkl}$ reciprocal lattice vector of the single crystal sample into the Laue condition, where $K–K_0 = G_{hkl}$. $G_{hkl}$ is perpendicular to the $hkl$ diffraction planes with modulus $|G_{hkl}| = 2\pi/d_{hkl}$, where $d_{hkl}$ is the diffraction plane spacing. $K_0$ and $K$ are the incident and scattered wave vectors with modulus $2\pi/\lambda$. $|K–K_0| = 4\pi\sin\theta/\lambda$, where $2\theta$ is the scattering angle. The scalar equivalent of the Laue condition is $\lambda = 2d\sin\theta_B$ (Braggs' Law). As pictured here $\omega = \theta$ and $\chi = 90°$, in which case the selected $G_{hkl}$ is parallel to the $\phi$ axis, which is nominally the surface normal direction.

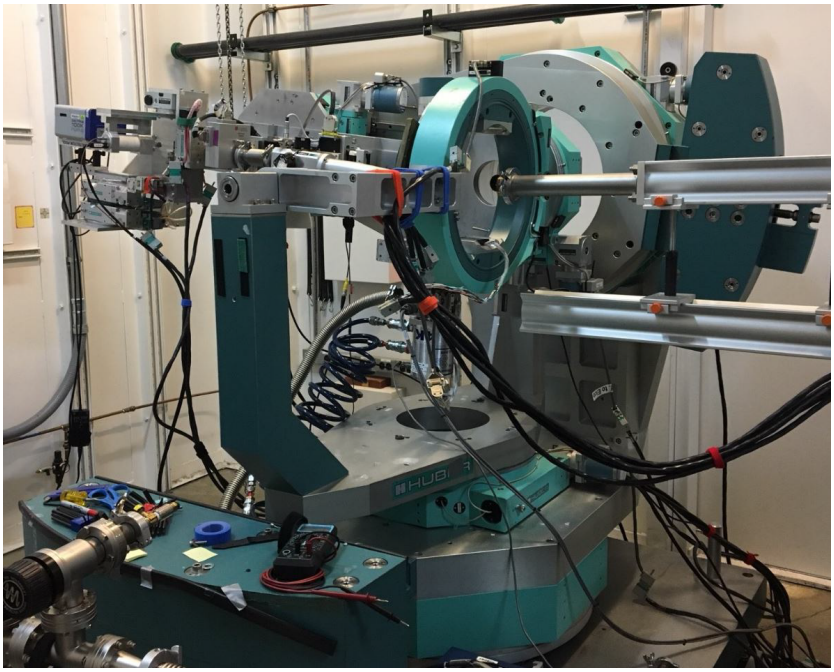




Figure 3: Photographs of a Huber 6-circle diffractometer, with a similar cryostat installed similar to the one proposed here. Instead of the beryllium dome pictured, we will have beryllium windows for x-ray transmission through the outer vacuum enclosure.

- **Denis Keane** is an NU Research Professor in Materials Science and Engineering and an expert on X-ray instrumentation and diffraction experiments. He is the Director of the DuPont-Northwestern-Dow Collaborative Access Team (DND-CAT) located at the Advanced Photon Source (APS) of Argonne National Laboratory (ANL).

## b. Facility Access

Planning for synchrotron experiments always faces the risk of not getting beam through the general user proposal process. Northwestern faculty are fortunate to have access to two facilities (DND- and LS-CAT) operated by NU at the Advanced Photon Source that provide dedicated access for NU faculty members.

### b..1 Jerome B. Cohen X-Ray Diffraction Facility

Northwestern's Evanston campus houses the Jerome B. Cohen X-Ray Diffraction Facility (http://www.xray.facilities.northwestern.edu). This facility is located in a neighboring building to the Odom lab, with interior-hallway access permitting easy moving of apparatus back and forth. The cryostat purchased for this proposal will be mounted to an existing 4-circle rotating anode setup in this facility, which will be readily available to this project.

### b..2 DuPont-Northwestern-Dow Collaborative Access Team Facility

The local facility is of course convenient, but there are advantages of much higher brightness, energy tunability and polarization for performing XRD experiments at a synchrotron. Northwestern is the operating partner of the DND-CAT facility at APS/ANL. (http://www.dnd.aps.anl.gov). This is a state-of-the-art x-ray facility located in Sector 5 of the APS and available to Northwestern faculty. That gives Northwestern faculty the unusual benefit of consistent access to x-ray beamlines. It should be noted that the National Science Foundation was one the funders of the instrumentation at this facility. In addition to DND, we will submit general user proposals for performing experiments at APS operated beamlines (sectors 4 and 6 ) that specialize in low-temperature single crystal diffraction under magnetic fields, such as the 4ID-D undulator station with its Huber 6-circle with cryostat down to 8K and electromagnet up to 600 gauss.

## c. Experiments

### c..1 Undoped Crystal Studies

For studying Ar crystals, the NU facility will have a sufficient flux to perform the XRD measurements of the crystal domain size. For studying domain size in Ne crystals, we expect to need the higher flux of DND-CAT. By using standard XRD techniques, we will have sensitivity to the various possible defect types: point, line, planar, and bulk. These defects can be observed and identified by the different effects they have on the coherent and diffuse X-ray scattering. Part of the research effort will be to quantify the level of sensitivity to defects, i.e. to what level of imperfections might be hiding in what appears to be a single crystal. We note that we have chosen dopant and host matrix species for maximum sensitivity, and that the higher x-ray flux at the APS will yield higher sensitivity.

### c..2 Doped Crystal Studies

There is will be no periodicity, so we cannot look for superlattice peaks. Pair distribution function (PDF) studies can be used to characterize the distribution of Cs dopants within the lattice. These studies require the use of the tunable synchrotron x-rays at DND-CAT. These studies can be done for an Ar host matrix, but we note they will be even more sensitive in the Ne matrix due to larger scattering contrast.

The APS 5-BM-D (Sector 5 - Bending Magnet D Branch Beamline) contains a dedicated facility for extended x-ray absorption fine structure (EXAFS). Briefly, EXAFS observes oscillations in either characteristic x-ray fluorescence yield or x-ray absorption while the exciting x-ray beam energy is scanned through an atomic absorption edge. Scattering of the ejected photoelectron from the surrounding atoms results in oscillations of the signal. We can do Cs K-edge or $L_3$-edge XAFS to determine the Cs nearest neighbor bonding distances and coordination and Cs oxidation state or unoccupied density of states above the Fermi edge. These quantities will be important in understanding whether we are achieving the goal of having the dopants in a small number of well-defined local configurations.

#### c..3 Impurity Studies

Setting an upper limit on magnetic impurities will be important for understanding observed the spin coherence times of the Cs dopants. Given the cryogenic environment, we expect that it will not be hard to keep these to a low level. Oxygen molecules are the primary concern here.

X-ray fluorescence (XRF) and x-ray absorption (XRA) provide powerful diagnostics to study impurities within the crystal. These techniques require the tunability of the synchrotron source. Briefly, they involve scanning the x-ray energy over the absorption edges of possible impurity atoms, and observing the signals which would arise if those impurities are present. Achievable sensitivity is at the ppm level, which would be better than what we need to obtain ms $T_2^*$ times.

## 5 Optical Measurements

In the initial stages of analyzing undoped crystal quality, x-ray analysis will be used in parallel with various optical analyses. Our ultimate goal is obtaining good $T_2^*$ values for a high dopant densities. So, once we are satisfied with what the x-ray analysis reveals about the quality of doped samples, the weight of the analysis will shift to laser manipulation and probing of the dopants.

Most previous optical absorption analysis of doped cryocrystals has been performed in transmission mode. Our novel proposal for single-crystal growth makes reflection-mode analysis more convenient to implement, where the crystal is grown on a reflective substrate. This difference of technique should not be significant.

### a. Undoped Crystal Analysis

Optical analysis of the crystal can aid at the early stages of the project, perhaps preceding or in parallel with the first x-ray analysis. Non-resonant optical polarization measurements have been used to characterize the domain size of birefringent microcrystallites of inert matrices. The rare-gas crystals considered here have a lowest-energy fcc phase which by symmetry is not birefringent, so when things are working correctly, this mode of analysis would become obsolete. However, under non-ideal deposition conditions, a significant fraction of hcp microcrystallites have been observed. Optical polarization measurements can be performed as an early-stage diagnostic as we are learning to grow purely fcc crystals. Similarly, Raman polarization measurements can identify the presence of hcp domains. The advantage of the Raman technique over the simpler polarization rotation technique is that it is insensitive to birefringence of the windows.

Once undoped crystals are consistently grown with mostly fcc phase, we will need to rely on the more sensitive and informative x-ray analysis to set limits on residual hcp microcrystallites and any defects between fcc domains. When we move on to doped samples, we do not expect either of the polarization analyses to be sufficiently sensitive to reveal information about the local configurations of the dopants. We will instead rely on x-ray techniques for an initial analysis, and then we will move on to the AMO techniques discussed below.

### b. Spectral Analysis

Atoms trapped in inert matrices generally exhibit multiple (broadened) absorption lines, associated with different dopant trapping sites [46, 15]. These absorption lines can be up to a few hundred nanometers separated from one another. Elimination of all but one of these absorption lines would be expected if we succeed at our goal of growing a single crystal with a single well-defined local configuration for the dopants. This is a topic which has not been studied previously. As we experiment with different growth techniques, the spectrum will be monitored for such an improvement.

### c. Optical Pumping Efficiency

As in the envisioned for an eventual EDM experiment, we will use optical pumping to obtain a spin-polarized sample. Following the techniques of other groups [12, 4], the spin state is pumped with strong circularly polarized light. The effectiveness of pumping will be analyzed by circular dichroism, whereby weak right-hand-circular and left-hand-circular probe beams are sent to the sample. The pump and probe wavelengths wavelengths will be the same, and these beams will be from the same laser. The relative absorption of the probe beams defines the circular dichroism, from which can be extracted the spin polarization of the dopants. The pumping beam will be modulated on and off while the circular dichroism is monitored. In vacuum, 100% polarization can easily be achieved, but for alkalis in matrices a 'good' value is $\sim 10\%$, where the relative short-coming has been attributed to both imperfect pumping and imperfect readout. Although we plan to do optical pumping analysis solely via absorption measurements, the vacuum enclosures will be equipped with perpendicular windows at the sample height, suitable for fluorescence measurements should those be desired.

### d. Longitudinal Spin Relaxation: $T_1$

The $T_1$ time is only important insofar as it sets an upper limit on $T_2^*$. But it provides a good first measurement to make sure that things are under control. A $T_1$ measurement is obtained by observing the decay of the circular dichroism signal as a function of time after initial optical pumping. For dopant densities of interest, $T_1$ is set by fluctuation of magnetic impurities, or if those are at a sufficiently low level, fluctuating nuclear spin directions for $I \neq 0$ isotopes in the host crystal. For higher dopant densities, $T_1$ would be set by interaction between dopants, but we would not want to operate in this regime. For high densities of Rb in neon polycrystals, $T_1$ was observed to be 0.6 s [19]. For Cs in solid helium, $T_1$ of order 1 s was observed [12].

### e. Ensemble Dephasing Time: $T_2^*$

Relatively simple free induction decay can be used to measure $T_2^*$. To gain further information about what type of inhomogeneities are limiting $T_2^*$, we will use depolarization spectroscopy. Both techniques, applied in that case to Rb trapped in a parahydrogen matrix, are described in [16], including the Supplementary Material.

Free induction decay analysis of $T_2^*$ can be performed either at low bias magnetic fields ($\sim 10$ Gauss), where the individual transitions are not resolved in the Fourier transform of the decay does, or at higher bias fields where the decay shows spectrally resolved beat notes. The low-field case proceeds as follows. The Cs is optically pumped, then Larmor precession is induced by an RF pulse. With the probe lasers aligned at an angle to the bias field, the observed circular dichroism will exhibit the decaying oscillations typical of free induction decay. A fit to the exponential decay yields $T_2^*$.

Depolarization spectroscopy is important for distinguishing between electrostatic and magnetic origins of inhomogeneous broadening. The idea is that, since electrostatic interactions are time-reversal-invariant,

they will yield identical shifts for time-reversed states. So, electrostatic and magnetic sources of inhomogeneous broadening ($T_2^*$ degradation) can be distinguished by comparing dephasing times of superpositions of time-reversed states (such as a superposition of $m_F = +1$ and $m_F = -1$) with other superpositions. Electrostatic interactions with the host matrix (which are time-reversal invariant) will degrade $T_2^*$ for the latter but not the former. (Thus the reason for depolarization spectroscopy. Superpositions of time-reversed states have no net magnetization and do not exhibit Larmor precession, so the free-induction decay technique described above cannot be used to obtain their dephasing time.)

In depolarization spectroscopy, the atoms are initially polarized by optical pumping. The circular dichroism signal is continuously monitored while an RF drive is swept across the one-photon $\Delta m_F \pm 1$ and two-photon $\Delta m_F \pm 2$ resonances. As the RF drive redistributes population at each subsequent resonance, a degradation of circular dichroism is observed, and the polarization is step-wise driven toward zero. Without inhomogeneities, the RF would cause sudden dichroism shifts at the narrow resonant frequencies, but atoms experiencing different perturbations lead to a broadened response. It has been observed for both Rb in parahydrogen polycrystals [16] and neon polycrystals [19] that the $\Delta m_F \pm 1$ and non-time-reversing $\Delta m_F \pm 2$ resonances had widths agreeing with free-induction decay measurements of the sample inhomogeneity. However, the two-photon $m_F = +1 \Longleftrightarrow -1$ resonances (in both lower and upper manifolds) were much narrower and limited by other factors. Thus, in both polycrystalline host matrices, the Rb electron dephasing time was determined to arise from inhomogeneous electrostatic interactions with the host matrix.

As described previously, there is an orders-of-magnitude gap between obtained $T_2^*$ and observed $T_1$ times for alkalis trapped in polycrystalline matrices. Before creating high-quality doped single crystal samples, we will confirm by depolarization spectroscopy that our degradation of $T_2^*$ is electrostatic in origin. As we learn to make high-quality crystals, it our conjecture that we will see dramatic improvements in the ensemble dephasing time $T_2^*$, and that we will be able to obtain long $T_2^*$ times with high enough dopant densities to motivate EDM experiments with more exotic dopant species, such as heavy polar molecules.

### f. Spin Coherence Time: $T_2$

For application to an EDM measurement, we care about the ensemble dephasing time $T_2^*$. Other applications such as magnetometry, which can use spin-echo techniques to cancel inhomogeneities, can have a poor $T_2^*$ so long as a good spin coherence time $T_2$ can be obtained.

We do not plan to measure $T_2$, since generally this quantity can always be made to approach $T_1$ with enough sophistication in the spin-echo techniques. However, if we find ourselves confused about the $T_2^*$ values we are seeing, it might be reassuring to confirm that good $T_2$ values can be obtained. In that case, we could use DC or AC spin-echo techniques, as have been used previously [30].

### g. Bleaching

Unlike in free space, alkali atoms trapped in inert matrices are known to 'bleach', that is to stop scattering photons after some number of cycles. This behavior is attributed to reconfiguration of the atom into a different trapping site, enabled by energy deposited inelastically by a photon. The number of cycles before bleaching varies from 10 [32] to $10^9$[30], depending on the dopant/host combination. In most cases previously studied, the sample could be reset to its unbleached state by application of narrowband or broadband light, sometimes called 'optical annealing'.

It is plausible that the problem of bleaching is worse for more polycrystalline samples. This topic has not been previously studied. As we approach our goal of single doped crystals with a single well-defined dopant trapping site, we will keep track of the bleaching properties.

## 6 Conclusion

Alkali atoms trapped in cryogenic polycrystals have shown the potential of doped cryocrystals to be powerful platforms for precision measurements, such as in searches for an electron EDM. However, the huge potential advantage of these systems can only be realized if a way is found to eliminate inhomogeneities of dopant trapping sites in polycrystals. Since it has long been known that large-volume *undoped* single crystals of rare gas atoms can be grown, it is highly plausible that a way can be found to grow large volumes of *doped* single crystals, with dopant densities high enough to be of interest for precision measurements. We have proposed a novel approach toward achieving this goal, and an x-ray and optical analysis program which will provide metrics along the way.

## 7 Relation to Other Funding

The work proposed here builds on prior expertise of our lab, but it has no overlap with other research grants. This proposal is directly synergistic with the Convening Award from the Moore Foundation, which will allow us to host a workshop in 2022 bringing together people to discuss using doped cryocrystals for fundamental physics research.

## 8 Broader Impacts

If successful, this work will have great significance for the development of ensemble quantum sensors. By eliminating the inhomogeneous broadening associated with polycrystalline hosts and allowing all the implanted species to be addressed simultaneously, doped single crystals could be used to create higher-sensitivity sensors for magnetic fields or other physics probed by electron spin degrees of freedom [4, 16, 17, 18]. All fundamental physics applications would be expected to benefit as well, for the reasons detailed above [3, 5, 6, 7, 8, 9, 10, 47].

This technology is also expected to have a payoff in the development of single-atom quantum sensors, which have gained interest for their potential for performing NMR and MRI measurements of single molecules [18]. Such an idea has been proposed using NV centers [48, 49, 50, 51, 52], but there one has to 'hunt' for an NV center with good properties, as one would need to do in a doped polycrystalline matrix. In a doped single crystal, it is expected that every dopant atom or molecule would be a good quantum sensor, thereby making this technology more viable for real-world applications.

The work proposed here could also be of significant consequence for the development of quantum simulators using atoms and molecules trapped in solid cryocrystals. These systems have been proposed for the study of many-body spin physics beyond what can be studied with other platforms for quantum simulation [53]. The required ensemble dephasing times are beyond what can be realized in polycrystalline samples, but could be enabled by single-crystal doped cryocrystals.

As for the more local impact, graduate students and postdocs will be trained in cryogenic techniques, laser spectroscopy techniques, and x-ray analysis techniques. The latter will be conducted in collaboration with our Materials Science collaborators, so the training will be a strongly interdisciplinary. The PI has made a habit of encorporating aspects of his research into his undergraduate and graduate teaching, and he plans to also do so with the work proposed here.

# 9 Results from Prior NSF Support, Grant No. PHY-1806861, 7/1/2018 - 8/31/2022

Grant: 'Telecom-Band Rotational Cooling of a Heavy Molecular Ion', $449,999

**Intellectual Merit.** Under previous NSF support, we had succeeded in rotationally cooling trapped $AlH^+$. During this grant cycle, we worked on a new species $SiO^+$. Like $AlH^+$, $SiO^+$ has a diagonal transition we use for broadband optical pumping. Unlike $AlH^+$, it has a low-lying intervening state, which we used to our advantage. $SiO^+$ is more challenging to optically pump than $AlH^+$, because of its heavier reduced mass. The resulting smaller rotational spacing causes more states to be thermally populated and requires better spectral filtering to avoid driving heating transitions. However, there are two significant advantages over $AlH^+$: the dominant isotopologue has zero nuclear spin (compare with $I = 5/2$ for $^27$Al and $I = 1/2$ for H), and the rotational intervals are accessible with convenient microwave sources.

With the support of this grant, we demonstrated high-fidelity optical pumping of $SiO^+$ to its ground rotational state (published in PRL.) We also demonstrated that a modified spectral filter mask could pump to narrow distributions of so-called super rotors states, having very high rotational energies (published in Nature Commun.) We used super rotor states to probe the molecular structure far from its equilibrium geometry. We also demonstrated a factor-of-several change in the chemical reaction rate with $H_2$, dependent on the $SiO^+$ rotational excitation. We are in the process of working with theorist Hua Guo to understand the quantum mechanics underlying these observations.

**Broader Impacts.** The ability to control and read out molecular states non-destructively will potentially lead to applications involving quantum-controlled chemistry and quantum information processing with trapped polar molecular ions. This state control, including up to super rotor states, has potential to impact on work of chemists studying structure and reactions of molecules. $SiO^+$ has a microwave interval which is in the convenient range accessible by cheap commercial sources. It also has only a few isotopologues each with small nuclear spin, which could be chosen for different applications. These properties together make it very likely to be a favorite species in future quantum information processing work with molecular ions.

In the process of doing this research, a diverse group of students worked in the laboratory on this project, including one non-binary student, one male URM undergrad, and one female URM graduate student. Carissa Skye received their MS degree and went on to the PhD program at University of Illinois, Urbana-Champaign. Patrick Stollenwerk received his Ph.D. at the end of this work and went on to a postdoc position at Argonne National Lab. Ivan Antonov completed his postdoc in our lab and went on to a permanent Research Scientist position at the Lebedev Institute.

**Publications Citing NSF Grant No. PHY-1806861**

1.'Super Rotor Excitation Effects on Chemical Reaction Rates,' S. Venkataramanababu, James B. Dragan, Ivan O. Antonov, and B.C. Odom. *In Preparation*

2.'Optical Pumping of Molecules with Low-Lying Electronic States,' James B. Dragan, Ivan O. Antonov, and B.C. Odom. *In Preparation*

3.'Precisely Spun Super Rotors,' I.O. Antonov, P.R. Stollenwerk, S. Venkataramanababu, A.P. de Lima Batista, A.G.S. de Oliveira-Filho, and B.C. Odom. Nature Commun. 12, 2201 (2021)

4.'Cooling of a Zero-Nuclear-Spin Molecular Ion to a Selected Rotational State,' P.R. Stollenwerk, I.O. Antonov, S. Venkataramanababu, Y.-W. Lin, and B.C. Odom. Phys. Rev. Lett. 125, 113201 (2020)

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## Plan for Data management

*Types of Data Expected.* Research Activities will result in experimental data that will be produced from measurements and observations for example in the form of ascii time series data or images, including measurements from cameras, transducers, and data acquisition computer hardware/software. Records of experimental procedures, settings, and conditions will be recorded in digitized laboratory notebooks. In many cases samples will be produced using novel techniques, and the details of the techniques and recipes used will be documented for sharing with interested parties. All significant findings will be published in papers in refereed journals or conference proceedings, presented at scientific conferences, and furnished to popular media as appropriate.

*Access to data* will be provided in the form of papers, proceedings, talks, and PhD theses on the research group website. Also notable research highlights, experimental developments, and news/press links will be included as appropriate. Detailed descriptions of novel devices will be provided in the publicly available PhD theses as they are completed. Significant findings will also be posted on the public arxiv.org server often before formal Journal publications are produced. Detailed data sharing will be available upon request to interested parties as data and results become available.

The policy for use of data and images that are provided on the website will be open access. For other shared data requested by interested parties, reference to the source of data would be requested if further dissemination was intended by the recipient.

*For data archiving*, e.g. the ascii data and image data will be stored electronically and backed up on a remote network hard drive and on the Cloud to safeguard against fire or water damage. Digitized lab notebooks will also be stored remotely to ensure preservation of techniques, procedures used, and the recorded state of equipment during measurements. The data will be conveniently organized by category and date produced and will correlate with dated pages of laboratory notebooks. The data is expected to be stored indefinitely.

## Postdoctoral Mentoring Plan

Postdoctoral researchers will acquire a variety of experience with precision measurement techniques, lasers and optics, ultra-high-vacuum, low temperature physics techniques, x-ray analysis techniques, and data acquisition and analysis. These interdisciplinary skills along with the expected scientific publications from this work will place the post-docs in a strong position for future employment in the workforce.

*Orientation.* Postdocs will receive a package of orientation materials that describe the activities in their home department, including information about national associations for postdocs [1-2]. For example, Northwestern University resources are available for help with professional or personal concerns. Should a suitable female or minority candidate be selected for the position, efforts will be undertaken to connect them with support groups for minorities in physics to help them integrate into the broader research community. For example, the Office of Postdoctoral Affairs (OPA) at Northwestern provides career development resources and training. Materials on responsible conduct of research, career opportunities, CV preparation, interview skills, manuscript preparation, and the delivery of oral presentations will be made available through a reading list (APS career pages, e.g. [3,4]).

*Mentoring opportunities.* The postdocs will be given the opportunity to gain experience with the supervision of undergraduates and graduate students in the lab. This can be a very valuable skill for either an academic or industrial career.

*Research seminars and meetings.* The postdocs will present a yearly research seminar to our department and a constructive critique will be provided. The post-docs will be encouraged to attend and present research at scientific conferences such as the American Physical Society meetings. This serves to develop their knowledge of current activity in the field of quantum information science and sensing. Attendance at these meetings facilitates the development of creativity and ideas for new research pursuits. It also builds essential scientific communication skills and encourages networking and collaboration which can lead to fruitful outcomes in science and their career.

*Inclusion in proposal preparation and discussions of research directions.* Significant academic freedom will be encouraged for the postdoctoral scholar so that their independent new original ideas can be developed in conjunction with the laboratory work on the proposed projects. As part of the NSF mission is to educate and train the next generation of scientists, this flexibility is highly beneficial for fostering the development of new research directions. The postdocs will be encouraged to participate in grant proposals to develop experience with the grant writing process, as well as to develop their original ideas or ideas in collaboration with PI.

*Career counseling.* The PI will aim to provide timely career counseling about job opportunities and prospects. In addition they will hold a semi-annual review for each postdoc to address and document the progress toward their career goals. The PI will make contacts with other individuals in the field on their behalf to help facilitate their future job prospects.

[1] http://www.nationalpostdoc.org/

[2] http://www.phds.org/

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